

Jiadong Lu

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EDUCATION

Zhejiang University

Master in Control Science and Engineering

- FASTLab. Advised by [Chao Xu](#) and [Yanjun Cao](#)

China

Sep. 2024 – Present

Shandong University

Bachelor in Control Science and Engineering

- GPA: 4.0/4.0 (Rank: 2/28)

China

Sep. 2020 – June 2024

RESEARCH INTERESTS

My research focuses on robotic perception, including ego-state estimation for individual robots and collaborative perception for multi-robot systems. My long-term research goal is developing accurate, real-time state estimation algorithms that enable robots to perceive themselves, understand their surroundings, and operate robustly.

PUBLICATIONS († denotes equal contribution)

[1] Clamped Non-Uniform B-Splines for Continuous-Time Odometry and Mapping

Jiadong Lu, Zhehan Li, Jiajun Lv, Chao Xu, Yanjun Cao. *In preparation*.

[2] Parallel Continuous-Time Relative Localization with Augmented Clamped Non-Uniform B-Splines [\[Paper\]](#)

Jiadong Lu[†], Zhehan Li[†], Tao Han, Miao Xu, Chao Xu, Yanjun Cao. *Submitted to IEEE Transactions on Robotics*.

[3] Hierarchical Cooperative Relative Localization via Closed-Form Solutions and Relative Kinematics [\[Paper\]](#)

Z. Li, J. Lu[†], Z. Wang[†], Z. Xun, C. Xu, F. Gao, Y. Cao. *Submitted to IEEE Transactions on Robotics*.

[4] Mr. Virgil: Learning Multi-robot Visual-range Relative Localization [\[Paper\]](#)

S. Wang, Z. Li, J. Lu, R. Xiong, Y. Cao, Y. Wang. *IROS, Oral*. 2025.

[5] Novel Design of Reconfigurable Tracked Robot with Geometry-Changing Tracks [\[Paper\]](#)

C. Xuan[†], J. Lu[†], Z. Tian, J. Li, M. Zhang, H. Xie, J. Qiu, C. Xu, Y. Cao. *IROS, Oral*. 2024.

RESEARCH EXPERIENCE

Continuous-Time LiDAR-Inertial-Visual Odometry and Mapping

- Developed a continuous-time odometry and mapping system based on clamped non-uniform B-splines, enabling real-time estimation while maintaining accuracy comparable to conventional B-splines.
- Proposed closed-form prior transport to resolve Schur-complement prior inconsistency induced by C-NUBS extension, enabling consistent long-term estimation.
- Derived dense continuous-time PGO that preserves full trajectory information with near-discrete-PGO efficiency.
- Manuscript [1] in preparation, targeting IEEE Robotics and Automation Letters.

Continuous-Time Multi-Robot Direct Relative Localization

- Introduced clamped non-uniform B-splines for state representation, eliminating the query-time latency of conventional B-spline formulations.
- Formulated a continuous-time factor graph for relative localization, enabling natural fusion of asynchronous multi-robot observations and online inter-robot time-offset estimation.
- Developed a robot-wise asynchronous block coordinate descent solver, enabling real-time optimization of multi-robot relative states.
- Paper [2] submitted to IEEE Transactions on Robotics (under review).

Hierarchical Multi-Robot Direct Relative Localization

- Designed an online, robust, and fast hierarchical initialization method, which can overcome over 90% outliers and obtain accurate states of all neighbors in a time window.
- Developed a real-time tightly-coupled sliding-window optimizer that fuses multi-robot IMU preintegration, UWB distances, and camera bearings under robocentric relative kinematics for NLOS scenarios.
- Paper [3] submitted to IEEE Transactions on Robotics (under review).

Learning Multi-robot Visual-range Relative Localization

- Co-developed an end-to-end multi-robot UWB-visual relative localization framework.
- Designed a multiplex graph attention network with self- and cross-edge message passing, coupled with differentiable Sinkhorn assignment and a PGO back-end for joint front-end-back-end training.
- Paper [4] accepted by IROS-2025 as an oral presentation.

Mechanical Design of A Reconfigurable Tracked Robot

- Designed a reconfigurable tracked robot with geometry-changing tracks, a passive tension-maintaining mechanism, and an end-wheel-driven actuation scheme, achieving improved mechanical efficiency.
- Paper [5] accepted by IROS-2024 as an oral presentation.

Reinforcement Learning of A Reconfigurable Tracked Robot

- Investigated PPO-based joint morphology-locomotion control in IsaacSim, introducing cross-attention between track morphology and locomotion states to significantly improve system performance.
- Manuscript in preparation, targeting IEEE Robotics and Automation Letters.

PROJECT EXPERIENCE

Air-Ground Cooperation without Global Information

Zhejiang, China

IROS 2025 EXPO | Team Leader

2025

- Live demo showcasing autonomous collaboration between RoFly, an amphibious quadrotor, and CubeTrack [5], a reconfigurable tracked vehicle, enabled by our relative localization systems [2–4] and a non-inertial MPC controller without global information.

HONOURS AND AWARDS

Presidential Scholarship Finalist

Top 1%

Shandong University

2023

Excellent Graduation Thesis

Top 5%

Shandong University

2024

First Prize, National College Student Intelligent Car Competition

Top 5%

Chinese Association of Automation

2022

Second Prize, National College Student Intelligent Car Competition

Top 10%

Chinese Association of Automation

2021

Advanced Individual in Innovation

Top 5%

Shandong University

2023

First Prize Scholarship

Top 5%

Shandong University

2022 & 2023

SKILLS

Programming: C++ (5+ years), Python (5+ years), MATLAB (4+ years).

Frameworks & Libraries: ROS (3+ years), Eigen, Ceres, PyTorch, OpenCV, Open3D.

Development & Deployment: Linux, IsaacSim, Git, Docker, SolidWorks, Fusion, LaTeX.